

Precommitted gdTAI V4 Training and Validation Protocol

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Precommitted gdTAI V4 Training and Validation Protocol

- **Protocol version:** 1.2
- **Frozen:** 2026-08-07
- **Amendments:** M4 CD4/Treg exclusion recall-cost preflight; M5 audited transformed-input contract and sensitivity-only sorted cohorts, both added before fitting
- **Status:** supervision required; model fitting has not started
- **Primary objective:** improve dataset-macro balanced F1 without relaxing the precommitted alpha-beta T-cell and NK false-positive guardrails

1. Purpose and governance

This protocol freezes the next gdTAI development cycle before any new fitting, threshold search, or model comparison. It responds to the independent methodology audit by replacing cell-level train/tune splits with nested cross-dataset evaluation, using expression-independent TCR labels, and moving every feature decision, calibration step, and threshold choice inside grouped resampling.

The current gdTAI v3 Round 14 model remains the promoted balanced default. Round 12 remains the validated high-purity fallback. The existing experimental V4 artifact is not promoted and must be archived before a new V4 candidate is written. No result from this protocol may be called independent external validation: all available positive cohorts have already influenced prior development or review.

Training is blocked until this document, its rendered PDF, and the preflight audit are reviewed. A second supervision gate is required after nested evaluation and before any full-atlas inference or model promotion.

2. Fixed analysis population and data contract

2.1 Input population

- The model accepts T/NK-filtered single-cell RNA-seq data only. It is not a pan-cell-type classifier.

- Input values must be raw non-negative integer counts, except for the registered HRA005041 object whose only matrix is already `log1p(CP10K)`. Raw inputs are transformed as `log1p(counts / total_counts_per_cell * 10000)` without densifying the full matrix; HRA005041 is consumed without a second normalization.
- HRA005041 is eligible only if a full sparse-matrix audit confirms finite, nonnegative values, no empty cells, and `abs(sum(expm1(X_i)) - 10000) <= 1e-4` for every cell. This is an input-representation exception, not permission to accept arbitrary normalized matrices; any other transformed input requires a new protocol version.
- The canonical development atlas is `high_speed_temp/Integrated_dataset/integrated.h5ad`, currently containing 3,705,306 cells after the approved no-TCR-gene source carve-out.
- The GSE144469 expression matrix must come from the 107,068-cell raw-count object at `data/datasets/GSE144469/raw/legacy_source/fastq/tnk_subset.h5ad`. Its cells must be joined to canonical metadata by `SRR + barcode`; the join must be 100% unique and hashes of both source files must remain unchanged.
- BALF_BLOOD_COPD is a reused cross-study development benchmark, not an untouched test set. Its count layer is the only permitted expression source.
- H5AD inputs are read-only. Split assignments, labels, matrices, and predictions are written as separate tables or caches.

2.2 Required metadata

Every development row must provide a stable cell ID, `source_gse_id`, donor when available, `sample_id`, `library_id`, expression-state provenance, productive chain fields for TRA/TRB/TRG/TRD, canonical paired/any-chain flags, and doublet status when available. Clonotype IDs are required where TCR sequence permits them.

2.3 Primary truth labels

Primary labels are assigned from productive TCR metadata only. Expression, annotations, Phase-4 scores, and model predictions cannot change these labels.

Class	Fixed rule	Primary weight
<code>gdT_primary</code>	Productive paired TRG and TRD, with no productive TRA or TRB	1.0
<code>abT_primary</code>	Productive paired TRA and TRB, with no productive TRG or TRD	1.0
<code>dual_or_ambiguous</code>	Evidence for both receptor classes, conflicting flags, or incomplete contradictory metadata	Excluded
<code>single_abT_weak</code>	Productive TRA or TRB, no productive TRG/TRD	0.5, training only
<code>sorted_gdT_sensitivity</code>	GDT_2020AUG_woCOV, MalteGDT, or GDTlung sorted cells without primary TCR truth	Sensitivity only
<code>gdT_silver</code>	Project silver rule, including isolated productive TRD evidence	Sensitivity only

All three sorted cohorts are sensitivity-only. GDTlung2023july_7p remains sensitivity-only because of its suboptimal library quality; GDT_2020AUG_woCOV and MalteGDT are sensitivity-only because they cover only 82.7% and 41.6% of the frozen feature universe. Silver and sorted cells never enter feature selection, either model stage, calibration, threshold selection, or promotion metrics. TCR-confirmed CD4-like or Treg-like cells remain positive truth and count as false negatives if a post-model exclusion rejects them.

2.4 Dataset roles

- Primary nested evaluation sources: `HRA005041` , `GSE144469` , and `BALF_BLOOD_COPD` .
- Sorted sensitivity sources: `GDT_2020AUG_woCOV` , `MalteGDT` , and `GDTlung2023july_7p` .
- Negative stress source: `GSE254249` .
- Frozen extension negative-control cohorts: `GSE114724` , `GSE121636_GSE121637` , `GSE159251` , `GSE292700` , `GSE294273_GSE294274` , `GSE296954` , and `GSE315928` .
- `GSE169246` is a reduced-feature schema-sensitivity cohort unless its feature coverage is repaired and separately approved. It cannot contribute a primary guardrail in its current form.

The extension cohorts contain no gdT-gold or gdT-silver cells under project TCR rules. They can measure paired/single alpha-beta T-cell and strict-NK false-positive behavior, but not recall, F1, ROC-AUC, or PR-AUC.

3. Fixed feature policy

3.1 Exclusions

The classifier cannot use Phase-4 TRD, TRAB, or TRD-minus-TRAB scores; source IDs; sample, donor, or library IDs; annotations; UMAP or latent coordinates; TCR metadata; or doublet labels at inference. B-cell and myeloid genes, annotations, and program scores are excluded from training and inference. B/myeloid annotations may be reported only as out-of-scope contamination checks after predictions are frozen.

3.2 Gene universe

The feature universe is frozen at **197 genes**, below the 220-gene protocol cap and the user's absolute 300-gene limit. All 197 were confirmed present in the canonical atlas, `BALF_BLOOD_COPD`, and the `GSE144469` raw-count object on 2026-08-07.

- **153 individual TCR genes:** the cross-input intersection listed in Appendix A.
- **14 T-lineage genes:** `CD3D` , `CD3E` , `CD3G` , `CD247` , `CD2` , `LCK` , `LAT` , `TRAT1` , `CD5` , `CD6` , `THEMIS` , `BCL11B` , `TCF7` , `LEF1` .
- **10 CD4/Treg-context genes:** `CD4` , `IL7R` , `CCR7` , `SELL` , `LTB` , `MAL` , `FOXP3` , `IL2RA` , `CTLA4` , `TIGIT` .
- **14 NK/cytotoxic-context genes:** `NKG7` , `GNLY` , `KLRD1` , `TYROBP` , `FCER1G` , `FCGR3A` , `KLRC1` , `KLRF1` , `PRF1` , `GZMB` , `CTSW` , `CCL5` , `EOMES` , `TBX21` .
- **6 T-cell-state genes:** `CD8A` , `CD8B` , `KLRB1` , `ZNF683` , `RUNX3` , `IKZF2` .

TRG genes are retained as individual weak evidence but receive no privileged rule or hard threshold. NK/cytotoxic genes are contextual features, not NK-specific vetoes, because bona fide gdT cells can express them strongly.

Inside each training fold, a gene is removed only if absent, constant, or detected in fewer than 0.05% of that fold's cells. No outcome-driven DEG screen is permitted before or outside the folds. Missing retained genes at inference are zero-filled only after coverage checks; the model must abstain when coverage is inadequate.

3.3 Deterministic derived features

Derived features are restricted to detection counts for the TRA, TRB, TRG, and TRD gene families; unweighted means of the fixed T-lineage, NK-context, CD4-helper, and Treg panels; and the Stage-1 out-of-fold T-lineage probability. These are computed only from the 197 genes. They are not the legacy Phase-4 module scores and use no matched control-gene sampling. An individual-gene-only ablation is mandatory.

4. Model architecture and negative sampling

4.1 Stage 1: soft T-lineage gate

Stage 1 is an elastic-net logistic classifier that estimates T-lineage probability. It uses the non-TCR genes plus `TRAC`, `TRBC1`, `TRBC2`, `TRDC`, `TRGC1`, and `TRGC2`. TCR-defined alpha-beta and gamma-delta T cells are positive. NK negatives require an NK annotation, no productive TRA/TRB/TRG/TRD evidence, and no doublet flag. When two independent annotation fields are available they must agree; a single available NK annotation is retained at weight 0.5.

The NK training sample is dataset-balanced and fixed as 60% representative NK, 20% T-like hard NK, and 20% gdT-like/TRDC-positive-TRDV-negative NK. The hard strata alter sampling only; they do not create NK truth labels. Stage 1 must achieve at least 99% recall for TCR-defined gdT cells and 98% recall for productive alpha-beta T cells in every primary outer evaluation source at its inner-fold-selected operating point.

Stage 1 is soft: all T/NK-input cells continue to Stage 2, and the out-of-fold Stage-1 probability is a Stage-2 feature. No single marker can hard-remove a cell before Stage 2.

Stage-1 hyperparameters are frozen to:

- `C`: 0.03, 0.1, 0.3, 1.0
- `l1_ratio`: 0.0, 0.25, 0.5, 0.75
- solver: `saga`; maximum iterations: 5,000; tolerance: 1e-4

4.2 Stage 2: gdT classifier

Two candidate families are compared under identical folds and weights:

1. Elastic-net logistic regression using the 197 genes, allowed derived features, and Stage-1 out-of-fold probability. Grid: `C` = 0.01, 0.03, 0.1, 0.3, 1.0; `l1_ratio` = 0.0, 0.25, 0.5, 0.75, 1.0; `saga`, 5,000 iterations, tolerance 1e-4.
2. Histogram gradient boosting with `learning_rate` = 0.03 or 0.07, `max_leaf_nodes` = 7 or 15, `min_samples_leaf` = 100 or 500, `l2_regularization` = 1 or 10, `max_iter` = 250, and early stopping disabled so validation folds cannot leak into fitting.

No additional candidate family or grid point may be introduced after results are seen. Any later change starts a separately named, separately precommitted experiment.

4.3 Weighting

Training weights are the product of truth reliability, inverse dataset frequency within class, and inverse class frequency within dataset. Weights are normalized to mean 1 in each training fold. No cell is duplicated. Donor/library groups cannot be split between train and validation. Single-chain weak negatives use the fixed reliability weight in Section 2.3; sorted and silver cells have zero training weight.

5. Nested evaluation and operating points

5.1 Outer and inner splits

- Outer evaluation is leave-one-dataset-out across HRA005041, GSE144469, and BALF_BLOOD_COPD. The held-out source is absent from both model stages, feature filtering, calibration, and threshold selection.
- Each outer-training partition uses three-fold `StratifiedGroupKFold` for tuning.
- The group key is donor when available, otherwise sample, otherwise library. If donor is unavailable and a clonotype spans multiple fallback groups, the entire clonotype is assigned to one fold.

- Split manifests are generated once from a fixed seed (`20260807`), checksum-pinned, and reused by every candidate and comparator.
- Every feature filter, scaler, derived feature, Stage-1 prediction, hyperparameter decision, sigmoid calibrator, and operating threshold is fitted using inner-training data only. Isotonic calibration is sensitivity-only and cannot select a model.

5.2 Threshold selection

Threshold candidates are the sorted unique inner out-of-fold calibrated probabilities plus 0 and 1. Ties are resolved by lower strict-NK FPR, then lower paired-abT FPR, then fewer retained genes, then lexical candidate ID.

- **Balanced mode:** maximize dataset-macro F1 subject to macro recall at least 0.80, paired-abT FPR at most 0.20%, and strict-NK FPR at most 1.00%.
- **High-purity mode:** maximize dataset-macro F0.5 subject to macro recall at least 0.70, paired-abT FPR at most 0.10%, and strict-NK FPR at most 0.50%.

If no threshold satisfies a mode's guardrails, that mode fails for the candidate; the constraints cannot be relaxed. The final deployment thresholds are selected from cross-dataset out-of-fold predictions only after the candidate family is frozen. The resulting deployment thresholds are not reported as independently validated estimates.

5.3 CD4/Treg exclusions and VDJ-aware rescue

Both RNA operating modes reject coherent CD4-helper-like and Treg-like predictions after probability thresholding:

- CD4-helper exclusion: `CD4 > 0` , at least two of `IL7R` , `CCR7` , `SELL` , `LTB` , `MAL` are detected, and their six-gene mean is at least 0.5.
- Treg exclusion: `FOXP3 > 0` , at least one of `IL2RA` , `CTLA4` , `TIGIT` is detected, and their four-gene mean is at least 0.5.

These rules operate on log1p(CP10K) values and never alter truth labels. A separate, optional VDJ-aware output may rescue rejected cells with productive paired TRG/TRD and no productive TRA/TRB. Rescued calls are reported separately and never enter TCR-defined validation metrics, avoiding use of the truth fields to inflate model performance.

Before Step 2, the preflight must apply both exclusions directly to every `gdT_primary` cell and report, for each primary source and the unweighted source-macro summary: total primary positives; CD4-helper-only, Treg-only, overlapping, and union exclusion counts and proportions; Wilson 95% intervals; donor/library concentration; and the **exclusion-imposed recall ceiling** $1 - \text{union_excluded} / \text{gdT_primary}$. This ceiling is the best recall any RNA classifier can achieve after the fixed exclusions, even if its pre-exclusion recall were perfect. Report the corresponding absolute and percentage-point margin above the 0.80 macro-recall selection guardrail and the 0.70 per-source promotion floor. Sorted sensitivity cells and silver cells receive the same descriptive audit but cannot alter feasibility or selection.

Preflight fails and training remains blocked if the source-macro ceiling is below 0.80 or any primary source ceiling is below 0.70. A ceiling within 0.05 of either applicable floor is a supervision warning requiring explicit approval before Step 2. The cutoffs in this section are immutable once Step 2 begins or any Step-2 result is generated. They must never be re-tuned in response to nested-evaluation results. A later cutoff change requires a new protocol version and commit, a fresh preflight, and a new supervision approval; it cannot be used to rescue or reinterpret the current V4 experiment.

6. Comparators, metrics, and statistics

6.1 Fair retrained comparators

Every comparator uses the same outer and inner partitions:

1. Legacy TRD-minus-TRAB score with a fold-selected threshold.
2. Compact logistic model using **TRDC** , **TRDV1** , **TRDV2** , **TRDV3** , **TRAC** , **TRBC1** , and **TRBC2** .
3. V2-like single-stage elastic-net logistic regression using the fixed individual TCR genes only.

Frozen gdTAI V2 high-purity, V3 Round 12, and V3 Round 14 are descriptive reference profiles only because they cannot be fairly retrained from one coherent recorded build.

6.2 Primary reporting

For each outer dataset and the unweighted dataset macro-average, report ROC-AUC, PR-AUC, precision, recall, specificity, F1, F0.5, balanced accuracy, MCC, confusion counts, paired-abT FPR, strict-NK FPR, and calibration intercept/slope, Brier score, and expected calibration error. Also report silver recall, sorted-source sensitivity, CD4/Treg exclusion losses, TRDC-positive/TRDV-negative behavior, feature coverage, abstention, and PPV/FDR at gdT prevalences of 0.1%, 0.5%, 1%, 2%, and 5%.

Use 2,000 paired hierarchical bootstrap replicates, resampling datasets and then donors, or libraries when donor is absent. Report percentile 95% confidence intervals for each metric and paired candidate-minus-comparator differences. Dataset-level and worst-dataset results are primary; pooled cell-level results are secondary.

6.3 Promotion rule

V4 is promoted only if all conditions hold for the balanced mode:

1. Dataset-macro F1 exceeds the strongest fairly retrained comparator by at least 0.01.
2. The 95% paired hierarchical-bootstrap lower confidence bound for that F1 difference is greater than 0.
3. Recall is at least 0.70 in every primary outer dataset.
4. All balanced paired-abT and strict-NK guardrails pass in every primary outer dataset and in the seven complete-feature extension negative-control cohorts.
5. Feature coefficients or permutation ranks are directionally stable in at least two of three outer folds, and no single dataset accounts for more than 50% of aggregate feature importance.
6. The release bundle passes semantic and checksum consistency tests.

Failure of any condition retains V3 Round 14 as the promoted default and publishes V4 as an experimental negative result. High-purity mode may be distributed only when its own guardrails pass; it cannot rescue a failed balanced-mode promotion.

7. Full-atlas inference and release contract

Whole-atlas inference is blocked until the nested report is reviewed and promotion is explicitly authorized. After authorization, score the current 3,705,306-cell atlas once with the frozen final fit. Each output row must contain:

- calibrated Stage-1 T-lineage probability
- calibrated Stage-2 gdT probability
- balanced and high-purity RNA calls
- CD4-helper and Treg exclusion flags
- feature coverage and schema-abstention flag
- optional VDJ-aware rescued call
- deterministic decision reason
- model ID, artifact SHA-256, Git commit, and split-manifest ID

The model abstains when any critical gene (`CD3D` , `CD3E` , `TRAC` , `TRBC1` , `TRBC2` , `TRDC` , `TRDV1` , `TRDV2` , `TRDV3`) is absent or when retained model-gene coverage is below 90%. An immutable release bundle must include the model, feature order, coefficients or importances, normalizer, calibrators, thresholds, input hashes, split IDs, environment lock, command, random seeds, metrics, model card, inference example, and a manifest whose semantics agree with the serialized artifact and registry.

8. Step gates and required artifacts

Step 0: protocol freeze (this document)

- Commit this Markdown protocol, reader HTML, and PDF before any fitting.
- Email the PDF and commit SHA to Likai with supervision status and exact paths.
- Stop. No training or threshold search is authorized by completion of Step 0.

Step 1: preflight and split freeze

- Archive the existing experimental V4 artifact without overwriting it.
- Verify the frozen raw-count or audited transformed-input state, hashes, required metadata, label conflicts, class counts, feature coverage, and the GSE144469 `SRR + barcode` join.
- Quantify the fixed CD4/Treg exclusions on primary positives and report their per-source and source-macro recall ceilings, guardrail margins, overlap, and donor/library concentration using the rules in Section 5.3.
- Emit checksum-pinned cell-label and grouped-split manifests plus a preflight report.
- Email Likai and stop for approval.

Step 2: nested evaluation

- Run both stages, ablations, and fair comparators under the frozen folds.
- Emit per-fold predictions, metrics, calibration, feature stability, bootstrap results, extension negative controls, HTML, and a print-safe PDF.
- Email Likai and stop for model-selection review.

Step 3: final fit and decision

- Only after approval, fit the selected family on all development data, select final cross-validated operating thresholds, run stress cohorts, and apply the promotion rule.
- Publish either an immutable promoted V4 release or an experimental failure record.
- Whole-atlas inference requires separate explicit approval.

9. Mandatory automated checks

Before any candidate can run, tests must prove:

1. Every fitting input is either raw counts or the uniquely registered HRA005041 `log1p(CP10K)` matrix passing the full per-cell inverse-library-size audit; all inputs remain unchanged by size, mtime, and SHA-256.
2. Primary positive and negative labels are nonempty, mutually exclusive, and independent of expression and annotations; silver cells are absent from fitting and selection.
3. GSE144469 maps 107,068 of 107,068 cells uniquely by `SRR + barcode` .
4. The gene universe contains exactly 197 genes, no Phase-4 score, no metadata feature, and no B/myeloid feature; every fold retains at most 220 genes.
5. No donor, fallback sample/library, or protected clonotype crosses a fold boundary.
6. Feature filtering, Stage-1 probabilities, scaling, calibration, model selection, and thresholds are fold-local.
7. CD4/Treg exclusions and VDJ rescue are deterministic; rescue never changes the RNA metric frame; and preflight emits exact per-source/union counts, recall ceilings, and guardrail margins for primary positives.

8. The recorded CD4/Treg rule hash matches the Step-2 implementation and cannot change after nested evaluation begins.
9. Missing critical genes or less than 90% coverage causes abstention.
10. Two runs with the same seed produce identical split manifests, predictions, and model hashes, or a documented deterministic numerical tolerance where the estimator cannot be bitwise stable.
11. Serialized model, manifest, registry, report, and example inference outputs agree on feature order, thresholds, version, normalization, and checksums.

Appendix A. Frozen individual TCR genes (153)

TRAC , TRAJ10 , TRAJ12 , TRAJ13 , TRAJ16 , TRAJ18 , TRAJ22 , TRAJ23 , TRAJ27 , TRAJ28 , TRAJ32 , TRAJ34 , TRAJ37 , TRAJ38 , TRAJ39 , TRAJ42 , TRAJ43 , TRAJ44 , TRAJ45 , TRAJ46 , TRAJ47 , TRAJ49 , TRAJ5 , TRAJ6 , TRAJ9 , TRAV1-1 , TRAV1-2 , TRAV10 , TRAV12-1 , TRAV12-2 , TRAV12-3 , TRAV13-1 , TRAV13-2 , TRAV14DV4 , TRAV15 , TRAV16 , TRAV17 , TRAV18 , TRAV19 , TRAV2 , TRAV20 , TRAV21 , TRAV22 , TRAV23DV6 , TRAV24 , TRAV25 , TRAV26-1 , TRAV26-2 , TRAV27 , TRAV29DV5 , TRAV3 , TRAV30 , TRAV34 , TRAV35 , TRAV36DV7 , TRAV38-1 , TRAV38-2DV8 , TRAV39 , TRAV4 , TRAV40 , TRAV41 , TRAV5 , TRAV6 , TRAV8-1 , TRAV8-2 , TRAV8-3 , TRAV8-4 , TRAV8-5 , TRAV8-6 , TRAV9-2 , TRBC1 , TRBC2 , TRBJ1-1 , TRBJ1-2 , TRBJ1-4 , TRBJ1-5 , TRBJ1-6 , TRBJ2-1 , TRBJ2-2 , TRBJ2-2P , TRBJ2-3 , TRBJ2-4 , TRBJ2-5 , TRBJ2-6 , TRBJ2-7 , TRBV1 , TRBV10-1 , TRBV10-2 , TRBV10-3 , TRBV11-1 , TRBV11-2 , TRBV11-3 , TRBV12-2 , TRBV12-3 , TRBV12-4 , TRBV12-5 , TRBV13 , TRBV14 , TRBV15 , TRBV16 , TRBV18 , TRBV19 , TRBV2 , TRBV20-1 , TRBV21-1 , TRBV23-1 , TRBV24-1 , TRBV25-1 , TRBV27 , TRBV28 , TRBV29-1 , TRBV3-1 , TRBV30 , TRBV4-1 , TRBV4-2 , TRBV5-1 , TRBV5-3 , TRBV5-4 , TRBV5-5 , TRBV5-6 , TRBV6-1 , TRBV6-2 , TRBV6-4 , TRBV6-5 , TRBV6-6 , TRBV7-2 , TRBV7-3 , TRBV7-4 , TRBV7-5 , TRBV7-6 , TRBV7-7 , TRBV7-9 , TRBV9 , TRDC , TRDJ1 , TRDJ2 , TRDJ3 , TRDV1 , TRDV2 , TRDV3 , TRGC1 , TRGC2 , TRGV1 , TRGV10 , TRGV11 , TRGV2 , TRGV3 , TRGV4 , TRGV5 , TRGV5P , TRGV7 , TRGV8 , TRGV9 .

Appendix B. Precommit record

- Canonical environment: `rapids_sc_py310`
- Random seed: `20260807`
- Resource design limit: 80 CPU cores and 400 GB RAM; the active runbook permits up to 800 GB, but the workflow cannot require it.
- Current promoted comparator: gdTAI v3 Round 14, threshold 0.936
- High-purity fallback: gdTAI v3 Round 12, threshold 0.5
- Existing experimental V4: not promoted; archive before Step 1
- CD4/Treg exclusion values: fixed since protocol v1.1; no retuning after Step 2 begins
- HRA005041 transformed-input exception and sensitivity-only sorted roles: fixed in protocol v1.2 before fitting
- No training, calibration, threshold search, full-atlas inference, or model promotion was performed as part of Step 0.